

CH5 Special Discrete Distributions

Bernoulli and binomial R.V.

Definition

X is a Bernoulli trial (or Bernoulli random variable) with parameter p if sample space $S=\{s, f\}$, where s is called a success and f a failure and $X(s)=1$, $X(f)=0$.

The probability mass function of X is

$$p(x) = \begin{cases} p & x = 1 \\ 1 - p \equiv q & x = 0 \\ 0 & \text{otherwise} \end{cases}$$

$$E(X) = 0 \times P(X = 0) + 1 \times P(X = 1) = p$$

$$E(X^2) = 0^2 \times P(X = 0) + 1^2 \times P(X = 1) = p$$

$$Var(X) = E(X^2) - (E(X))^2 = p - p^2 = p(1 - p)$$

Definition

$$X \sim B(n, p)$$

X : The number of success

n : Perform n indeoendently event

p : The probability of success

The p.m.f. of $X \sim B(n, p)$

$$p(x) = P(X = x) = \begin{cases} \binom{n}{x} p^x (1 - p)^{n-x} & \text{if } x = 0, 1, 2, \dots, n \\ 0 & \text{elsewhere.} \end{cases}$$